

CS5099 - Dissertation Proposal

AI-based Anomaly Detection in Developer Access Control

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In modern software development, maintaining secure access control for developers is crucial to preventing unauthorised access, insider threats, and supply chain attacks. The frequency of major cyber attacks occurring due to over-privileged software developers has increased recently, including the developer access theft in the LastPass attack in 2022, where personal details and encrypted passwords were stolen [1].

It is common for software developers to have enhanced-access admin privileges, due to their continuous integration/continuous deployment style schemes, and very regularly developers will retain their access rights after they have finished with that feature. It is also very common for developers to be over-privileged, leading to the creation of dangerous back doors and access to the entirety of a system and database of a software.

Many solutions exist to this problem, such as Role Based Access Control, but constantly managing the roles that developers have and reassessing their permissions is not a valuable use of time and resources for business owners and not in the best interests of the developers themselves. Therefore, applying Machine Learning techniques to analyse developer activity and detect irregular behaviour, a proof-of-concept solution can be designed to help mitigate security threats and enhance access control mechanisms.

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The expected outcome of this project could involve a dataset of GitHub developer activity logs. A trained machine learning model may be developed to identify anomalous developer behavior, focusing on detecting irregular access patterns that could indicate security threats. The project may look towards producing a proof-of-concept anomaly detection system capable of monitoring developer activity in GitHub repositories. Additionally, a research paper will be written to document the methodology, findings, and performance evaluation of the model, highlighting its effectiveness in improving access control security.

Beyond the technical contributions, this project may enhance software security by providing an automated mechanism for detecting developer-related risks. By incorporating AI-driven anomaly detection into access control systems, organizations can better protect their repositories from insider threats, compromised credentials, and unauthorized modifications. The findings of this study may contribute to the growing field of AI-driven cybersecurity research while also providing valuable insights into developer behavior analytics in open-source and enterprise repositories.

[1] Dotan, Noam. "How Was LastPass Compromised?? Software Supply Chain Attack Tips." *Legit Security*, 29 August 2022, <https://www.legitsecurity.com/blog/lastpass-software-supply-chain-attack>. Accessed 12 February 2025.